Agent Instructions: Regression-Based Relative Valuation in Excel

Oct 28, 2026 · @Tommaso

1. Objective and deliverables

You are a valuation analyst agent. Your task: regress a valuation multiple on its fundamentals across the provided sample of companies, use the fitted equation to compute each company's predicted ("fair") multiple, and classify each company as undervalued, fairly valued or overvalued relative to the sample.

Regression scope. Work autonomously, knowing the 20 sample companies (the company list). Build a market-wide dataset of European listed companies in the same format as the professor's Japan example (section 1A), run the regressions on that whole market, and apply the fitted equations to each of the 20 sample companies. The final output is one valuation multiple per sample company (`valuation_multiples.csv`). Regressions on single clusters are optional cross-checks only: a single industry rarely has the 150+ companies a market-wide regression needs.

Course alignment. Relative valuation can be done relative to a sector (comparables) or to the entire market using regressions (Lecture 5, slide 40). The course shows both sector regressions (telecoms, slide 17; European banks, slide 29; European retailers, slide 38) and market-wide regressions (P/BV, slide 33; P/S, slide 39). The best estimate of value comes from one multiple (slide 41). Rules in this document that are not from the slides are marked as our rules.

Follow this document exactly. When it gives a rule, apply the rule; do not substitute your own judgment. When something is not covered, follow section 7 (stop conditions) and report it.

You must deliver exactly four files: the two below, plus DATA_BASE_EUROPE.xlsx and valuation_multiples.csv (section 1A, steps 2 and 7):

1. **`regression_valuation.xlsx`** — a workbook built to the specification in section 3. All statistics and valuations must be live Excel formulas referencing the data, not pasted values, so a human can audit and change any input.
2. **`valuation_summary.md`** — a written report following the template in section 6.

Definitions used throughout this document:

Term	Meaning
Y	The dependent variable: the valuation multiple (default P/E)
X variables	The independent variables: the fundamentals (default expected growth, beta, payout ratio)
k	Number of X variables in the model
n	Number of companies used in the regression after cleaning
df	Residual degrees of freedom = n − k − 1
Predicted multiple	Intercept + Σ (coefficient × company's X value)
Residual	Actual multiple − predicted multiple
SEy	Standard error of the regression (typical size of a residual)
Standardized residual	Residual ÷ SEy

1A. Building the dataset (Japan-example format)

Build a dataset of the European market in the same format as the professor's Japan example (`DATA_BASE.xlsx`: 3,528 Japanese companies, one row per company, one column per multiple or fundamental), regress the multiples on the fundamentals across the whole market, and apply the equations to the 20 sample companies.

Why market-wide. Relative valuation can be relative to a sector or to the entire market using regressions (Lecture 5, slide 40); the course's market regressions are estimated for Europe as a whole (slides 33 and 39), and the Japan example covers the whole Japanese market. All 20 sample companies are listed in Western Europe or the UK, so the European market is their reference market.

Step 1 – Universe

- Start from the constituents of the MSCI Europe index (about 400 large and mid caps), taken from the latest public holdings file of an ETF that tracks it; link the file. If time allows, extend to MSCI Europe IMI (adds small caps). More companies give more stable

coefficients.

- Make sure all 20 sample companies are in the dataset; add any that are missing and set `is_sample` = TRUE for them.
- Exclude banks, insurers, other financials and REITs (our rule): their multiples follow different drivers, which is why the course treats banks in a separate regression (Lecture 5, slide 29).
- Keep loss-making companies in the file: they still count for P/BV and P/S, and drop out of P/E automatically.

Step 2 – Columns (save as `DATA_BASE_EUROPE.xlsx`, sheet `Data`)

The first 18 columns copy the Japan file exactly, in the same order; the rest are added for this project. Write "NA" where a value does not exist, as the Japan file does. Never fill a gap with an estimate. All rates in decimals.

Column	Definition
Company Name	"Name (Exchange:Ticker)", as in the Japan file
beta	5-year regression beta against the local index
D/E	Total debt ÷ market value of equity
Price	Share price at `valuation_date`
Current PE	Price ÷ EPS of the last fiscal year (FY2025)
Trailing PE	Price ÷ EPS of the last 12 months
Forward PE	Price ÷ consensus EPS for next year; "NA" if not available (58% of the Japan file is NA)
PEG	Trailing PE ÷ (g × 100); "NA" if g ⩽ 0
PBV	Market value of equity ÷ book value of equity
PS	Market value of equity ÷ sales
EV/EBIT, EV/EBITDA, EV/Sales	EV = market value of equity + debt + minority interests − cash
Payout ratio	Dividends ÷ net income (DPS ÷ EPS)
Dividend Yield	DPS ÷ price

Column	Definition
g	**Fundamental growth = ROE × (1 − payout ratio)** (Lecture 3, slide 58). If ROE ⩽ 0, write "NA". If payout > 1, use payout = 1 (g = 0) and flag
Return on Equity	Net income ÷ book value of equity, last fiscal year
Net Profit Margin	Net income ÷ sales
ISIN, Country, Sector, Currency	Sector = GICS sector; Currency = the price currency
Market cap (EUR)	Converted at the `valuation_date` FX rate
Operating Margin	EBIT ÷ sales
Tax Rate	Tax expense ÷ profit before tax
ROIC	EBIT × (1 − tax rate) ÷ (book equity + debt − cash)
DFR	Debt ÷ (debt + market value of equity)
is_sample, cluster	TRUE and the cluster for the 20 sample companies; FALSE otherwise
source_urls	Linked sources for every figure (section 9)

Step 3 – Data sources, in order of preference

1. An export from Bloomberg, Thomson Eikon or S&P Capital IQ, if the user provides one (the Japan file's naming format matches a Capital IQ export). Ask the user for it before starting; if none arrives, continue with source 2.
2. stockanalysis.com statistics and financial-statement pages for each company (the source already used in the group workbook). For the 20 sample companies, cross-check against the group workbook and the annual reports.
3. Damodaran Online industry betas, only for the beta replacement in step 4.

Use one `valuation_date` for all prices and FX rates, and FY2025 financials for everyone. Convert per-share data to the price currency before computing any ratio.

Step 4 – Cleaning (our rules)

- For each model, drop rows with "NA" in Y or in any of its X variables.

- Drop non-positive multiples; for the P/E model, also drop P/E above 100.
- Replace a beta that is missing, negative or above 3 with a bottom-up beta (Lecture 2, slides 64–72) and flag it.
- Winsorize every Y and X at the 1st and 99th percentiles, so a few extreme firms cannot drive the line.
- Each model needs at least 150 companies after cleaning. If not, extend the universe (step 1) before running it.

Step 5 – Models to estimate

Model	Y	X variables	Basis
P/E	Trailing PE	g, beta, payout	P/E fundamentals (Lecture 5, slide 11)
P/BV	PBV	ROE, beta, payout	P/BV fundamentals and regression (slides 28, 33)
P/S	PS	Net profit margin, payout, g	P/S regression (slide 39)
EV/EBITDA	EV/EBITDA	g, DFR, tax rate, ROIC	Damodaran's European regression (not in the slides)

- Do not add g to the P/BV model: g is computed from ROE and payout, which are already in it, so it adds no new information.
- Run each model twice: market only, and with sector dummies (one 0/1 column per GICS sector except a base sector), the dummy technique of Lecture 5, slide 17. Keep the version with the higher adjusted R^2 and report both.
- Build each model in the workbook of sections 3–5, one set of sheets per model (`Regression_PE`, `Regression_PBV`, …). The LINEST rules apply with that model's X columns.

Warning from the Japan test. We ran these models on the Japan file. With g = ROE × (1 − payout), the P/E model gave a negative growth coefficient (−16.7, t = −5.4) and an R^2 of about 0.10 after cleaning, while P/BV on ROE, beta and payout gave an R^2 of about 0.35, with ROE strongly positive (t ≈ 36). Formula growth uses last year's ROE, which is a weak proxy for expected growth. Expect the same in Europe: Damodaran's January 2026 Western Europe regressions also rank P/BV first (R^2 42.3%), then EV/EBITDA (26.2%), P/E (20.1%) and EV/Sales (14.2%). Report such signs; do not hide them.

Step 6 – Choose the valuation multiple for each sample company

The professor's instructions: use the regression with the highest R^2 for the 2–3 relevant multiples (Agent Skills, Skill 3).

1. Rank the four models by adjusted R^2 on the European dataset.
2. For each sample company, a model is eligible only if its Y and X are meaningful for that company: no P/E model if EPS ⩽ 0 or P/E > 100; no P/BV if book equity ⩽ 0; no EV/EBITDA if EBITDA ⩽ 0.
3. The **valuation multiple** is the predicted multiple from the eligible model with the highest adjusted R^2. Also report the predicted multiples of the next two eligible models.
4. Cross-check: plug the company's fundamentals into Damodaran's January 2026 Western Europe & UK equation for the same multiple (pages.stern.nyu.edu/~adamodar, file MReg26). If the two predictions differ by more than 30%, flag it.

Step 7 – Output `valuation_multiples.csv`

One row per sample company:

`company, ticker, isin, cluster, chosen_model, adj_r2, actual_multiple, predicted_multiple, residual, std_residual, classification, implied_value_per_share, current_price, upside, second_model, second_predicted, third_model, third_predicted, damodaran_predicted, flags, source_urls`

Implied value per share = predicted multiple × the company's own metric (EPS, book value per share, sales per share), or, for EV/EBITDA, (predicted multiple × EBITDA − debt − minorities + cash) ÷ shares. Classification uses the bands in section 5.

2. Configuration and inputs

Use the configuration below unless the user's request overrides a value. Record the configuration you actually used in the `README` sheet and in the report.

Default configuration:

```
models: [PE, PBV, PS, EVEBITDA]   # section 1A, step 5
regression_scope: market_europe    # whole European dataset (Japan-example
format)
sector_dummies: test               # run with and without; keep higher
adjusted R2
universe: MSCI Europe              # extend to MSCI Europe IMI if time allows
exclude_sectors: [Banks, Insurance, Diversified Financials, REITs]
growth: fundamental                # g = ROE * (1 - payout), Lecture 3 slide
58
units_for_rates: decimal           # 0.15 means 15%
earnings_basis: FY2025             # never mix trailing and forward EPS
valuation_date: <one date for all prices and FX rates>
min_observations: 150              # per model after cleaning (our rule)
max_pe_for_estimation: 100         # (our rule)
winsorize: [0.01, 0.99]            # (our rule)
fair_band_se: 1.0                  # |std. residual| <= 1.0  -> Fairly valued
strong_band_se: 2.0                # |std. residual| > 2.0   -> Significantly
mis-valued
significance_level: 0.05
correlation_flag: 0.80
vif_warn: 5
vif_severe: 10
damodaran_crosscheck: MReg26, Western Europe & UK
```

Input data schema. One row per company. Column names below are canonical; map the user's headers to them and log the mapping.

Column	Type	Unit	Required	Notes
`company`	text	—	Yes	Name or ticker; must be unique
`isin`	text	—	Yes	Verified ISIN
`cluster`	text	—	Yes	Regression cluster from the company list
`is_sample`	TRUE/FALSE	—	Yes	TRUE for the 20 sample companies, FALSE for comparables
`price`	number	price currency	Yes	Share price at `valuation_date`

Column	Type	Unit	Required	Notes
`price_ccy`, `reporting_ccy`	text	—	Yes	Used for the currency conversion in step 4
`eps`	number	reporting currency	Yes	Earnings per share, same basis for all companies
`pe`	number	x	No	If missing, computed in step 5
`growth`	number	decimal	Yes	Expected EPS growth, usually analysts' 5-year estimate
`beta`	number	—	Yes	5-year regression beta, same source and window for all companies
`payout`	number	decimal	Yes	Dividends per share ÷ EPS
`roe`, `net_margin`	number	decimal	Only for PBV / PS	ROE = net income ÷ book equity; net margin = net income ÷ sales
`source_urls`	text	—	Yes	Link and access date for every figure (section 9)
`sector`	text	—	No	Carried through to the output

Unit normalization rule. For `growth` and `payout`: if the median of the column is greater than 1, the column is in percent. Divide every value by 100 and write one line in the `Log` sheet. Never normalize a single cell on its own.

Expected signs of the coefficients. These come from the fundamentals of the multiple (Gordon model: P/E = payout × (1+g) / (r − g)). They are used in the sign check in section 5.

Multiple (Y)	X variable	Expected sign (theory)	What the course regressions show
P/E	growth	+	+ (telecoms, Lecture 5 slide 17)

Multiple (Y)	X variable	Expected sign (theory)	What the course regressions show
P/E	beta	−	Not estimated in the slides
P/E	payout	+	Not estimated in the slides
P/BV	ROE	+	+ in US, Europe and EM (slide 33)
P/BV	growth	+	+ in US; not significant in Europe and EM (slide 33)
P/BV	payout	+	− in US; + in Europe and EM (slide 33)
P/BV	beta	−	+ in US and Europe; − in EM (slide 33)
P/S	net margin	+	+ in US, Europe and EM (slide 39)
P/S	growth	+	+ in US, Europe and EM (slide 39)
P/S	payout	+	− in US; + in Europe and EM (slide 39)

The theoretical signs come from dividing the dividend discount model by earnings, book value or sales (Lecture 5, slides 11, 28 and 37). The course's own data does not always agree, especially for beta and payout. A sign that differs from theory is therefore a flag to investigate, not an error in itself.

If the user configures a multiple or an X variable not listed here, ask the user for its expected sign before running the sign check.

3. Workbook specification

Build `regression_valuation.xlsx` with the sheets below, in this order. Formulas are given for the default model (Y = P/E; X = growth, beta, payout; k = 3). For another k, apply the general rule stated with each formula.

Sheet	Content
`README`	Configuration used, data source, date, column mapping, named cells `fair_band` and `strong_band`
`Raw_Data`	The input exactly as received. Never edit this sheet

Sheet	Content
`Clean_Data`	Companies used in the regression, after the cleaning in section 4
`Excluded`	Every removed company, with the rule that removed it
`Correlations`	Correlation matrix of Y and all X variables
`Regression`	Coefficient table and model statistics
`Diagnostics`	VIF per X variable and all flags from section 5
`Valuation`	Predicted multiple, residual, classification and implied price per company. Followed by a Robustness sheet when step 17 is triggered
`Targets`	Optional: companies to value that are not in the regression sample
`Log`	One line per decision, normalization, exclusion or warning

3.1 `Clean_Data` layout

Header in row 1, one company per row from row 2. The X columns must be **contiguous** and in the configured order.

Column	A	B	C	D	E	F	G	H
Field	company	price	eps	pe (Y)	growth (X1)	beta (X2)	payout (X3)	sector

Define these named ranges (last row = n + 1):

- `Y_range` = `Clean_Data!$D$2:$D$<last>`
- `X_range` = `Clean_Data!$E$2:$G$<last>`

3.2 `Regression` sheet

Use Excel's `LINEST(Y_range, X_range, TRUE, TRUE)`. Do not use the Analysis ToolPak dialog: its output is static values, not formulas.

LINEST output layout (critical). With `stats = TRUE`, LINEST returns 5 rows × (k + 1) columns. **Coefficients come back in REVERSE order of the X columns**, and the intercept is in the last column.

LINEST row	Col 1	Col 2	Col 3	Col 4 (= k + 1)
1	coef payout (X3)	coef beta (X2)	coef growth (X1)	intercept
2	SE payout	SE beta	SE growth	SE intercept
3	R^2	SEy	#N/A	#N/A
4	F	df	#N/A	#N/A
5	SS regression	SS residual	#N/A	#N/A

General rule: X variable number j (1 = first X column) is in LINEST column **k − j + 1**. The intercept is in column **k + 1**.

Read each value into its own labelled cell with `INDEX`, never by position in a spilled array. In the formulas below, `L` stands for `LINEST(Y_range, X_range, TRUE, TRUE)` written out in full.

Model statistics block (give each cell the name in the first column):

Named cell	Formula
`n_obs`	`=ROWS(Y_range)`
`k_vars`	`=COLUMNS(X_range)`
`R2`	`=INDEX(L,3,1)`
`Adj_R2`	`=1-(1-R2)*(n_obs-1)/(n_obs-k_vars-1)`
`SEy`	`=INDEX(L,3,2)`
`F_stat`	`=INDEX(L,4,1)`
`df_resid`	`=INDEX(L,4,2)`
`Sig_F`	`=F.DIST.RT(F_stat,k_vars,df_resid)`
`SS_reg`	`=INDEX(L,5,1)`
`SS_resid`	`=INDEX(L,5,2)`

Coefficient table (one row per term):

Variable	Coefficient	Std. error	t-stat	p-value	Lower 95%	Upper 95%	Expected sign	Sign OK
Intercept	`=INDEX(L,1,4)`	`=INDEX(L,2,4)`	coef ÷ SE	see below	see below	see below	—	—
growth	`=INDEX(L,1,3)`	`=INDEX(L,2,3)`	coef ÷ SE	see below	see below	see below	+	see below
beta	`=INDEX(L,1,2)`	`=INDEX(L,2,2)`	coef ÷ SE	see below	see below	see below	−	see below
payout	`=INDEX(L,1,1)`	`=INDEX(L,2,1)`	coef ÷ SE	see below	see below	see below	+	see below

Formulas for each row (B = coefficient, C = SE, D = t-stat, H = expected sign):

- t-stat: `=B/C`
- p-value: `=T.DIST.2T(ABS(D),df_resid)`
- Lower 95%: `=B-T.INV.2T(0.05,df_resid)*C`
- Upper 95%: `=B+T.INV.2T(0.05,df_resid)*C`
- Sign OK: `=IF(H="+",B>0,B<0)`

Name the coefficient cells `b0` (intercept), `b_growth`, `b_beta`, `b_payout`.

3.3 `Correlations` and `Diagnostics` sheets

- Correlation matrix: one cell per pair, `=CORREL(<column 1>,<column 2>)`, covering Y and every X. Name the X-only block (k × k) `corr_X`.
- VIF of X variable j: `=INDEX(MINVERSE(corr_X),j,j)` (the j-th diagonal element of the inverse correlation matrix).

3.4 `Valuation` sheet

One row per company in `Clean_Data`, same order. Columns:

Col	Field	Formula (row 2)
A	company	`=Clean_Data!A2`
B	actual multiple	`=Clean_Data!D2`
C	growth	`=Clean_Data!E2`
D	beta	`=Clean_Data!F2`
E	payout	`=Clean_Data!G2`
F	predicted multiple	`=b0+b_growth*C2+b_beta*D2+b_payout*E2`
G	residual	`=B2-F2`
H	standardized residual	`=G2/SEy`
I	% deviation from predicted	`=IF(F2>0,B2/F2-1,"n/a")`
J	implied price	`=IF(F2>0,F2*Clean_Data!C2,"n/a")`
K	current price	`=Clean_Data!B2`
L	implied upside	`=IF(F2>0,J2/K2-1,"n/a")`
M	classification	see below
N	rank (1 = most undervalued)	`=RANK.EQ(H2,$H$2:$H$<last>,1)`
O	flags	text written by the agent from section 5

Classification (column M):

```
=IF(F2<=0,"Not valued (negative prediction)",
 IF(H2<-strong_band,"Significantly undervalued",
 IF(H2<-fair_band,"Moderately undervalued",
 IF(H2<=fair_band,"Fairly valued",
 IF(H2<=strong_band,"Moderately overvalued","Significantly overvalued")))))
```

Add an XY scatter chart: actual multiple (y-axis) against predicted multiple (x-axis), with a 45° reference line. Points above the line trade above their fundamentals; points below trade below.

3.5 `Targets` sheet (optional)

Same columns as `Valuation`, for companies the user wants valued that are not in the regression sample. Add one extra column `extrapolation`: TRUE if any X value is below `MIN` or above `MAX` of that X column in `Clean_Data`.

3.6 Formula language

If you write the file programmatically (for example with openpyxl), write formulas in **English function names with comma separators**; Excel translates them for any locale. Only if you type formulas into an Italian-language Excel interface, use the local names and semicolons:

English	Italian
LINEST	REGR.LIN
INDEX	INDICE
ROWS / COLUMNS	RIGHE / COLONNE
T.DIST.2T	DISTRIB.T.2T
T.INV.2T	INV.T.2T
F.DIST.RT	DISTRIB.F.DS
CORREL	CORRELAZIONE
MINVERSE	MATR.INVERSA
RANK.EQ	RANGO.UG
IF / ABS	SE / ASS

4. Step-by-step procedure

Execute these steps in order. Do not skip a step. After each step, append one line to the `Log` sheet: step number, what you did, counts before and after.

1. **Build and load the dataset.** Build `DATA_BASE_EUROPE.xlsx` as in section 1A, steps 1–3. Copy it unchanged into `Raw_Data` and record the number of rows.
2. **Map columns.** Map the user's headers to the canonical names in section 2. Log the mapping. If a required column cannot be mapped with certainty, stop (section 7).
3. **Normalize units.** Apply the unit normalization rule of section 2 to `growth`, `payout`, `roe` and `net_margin`.
4. **Convert currencies.** When `reporting_ccy` differs from `price_ccy`, convert EPS (and DPS) into the price currency at the `valuation_date` FX rate before computing any multiple. Prices quoted in pence (GBX) are divided by 100 to get GBP. Log each FX rate with its source link. Examples in the sample: Vestas and NKT report in EUR but trade in DKK; ArcelorMittal and TotalEnergies report in USD but trade in EUR; Shell and BP report in USD but trade in GBX; Equinor reports in USD but trades in NOK.

5. **Compute Y.** In `Clean_Data`, column D is always the formula `=B2/C2` (price ÷ converted eps). If the input also has a `pe` column and it differs by more than 2%, log a warning. Use the computed value.
6. **Exclude invalid rows from the estimation.** Apply these rules in this order. Move each excluded company to `Excluded` with the rule number. A company excluded by one rule is not tested against the later ones.
   a. Duplicate `company` → keep the first occurrence, exclude the others.
   b. Any required field missing, non-numeric, or without a linked source → exclude.
   c. `eps` ⩽ 0 → exclude (P/E is not meaningful for loss-making companies).
   d. `price` ⩽ 0 or `payout` < 0 → exclude (data error).
   e. P/E > `max_pe_for_estimation` (earnings close to zero, e.g. BP in FY2025) → exclude from the estimation only. If it is a sample company, still list it in `Valuation` as "Not valued by regression (P/E above 100x)" and tell the user to value it with another multiple. (Our rule.)
7. **Replace unreliable betas.** If a beta is missing, negative, or above 2.5, replace it with a bottom-up beta: the unlevered beta of its industry relevered at the company's debt-to-equity ratio (Lecture 2, slides 64–72). Log the original and the replacement. In the sample this applies at least to Shell (missing), BP (−0.20) and Equinor (−0.66); also check TotalEnergies (0.09) and Verbund (0.19), which are implausibly low.
8. **Flag other implausible values (do not exclude).** Write a flag in `Diagnostics` and in column O of `Valuation` for: `growth` < −0.5 or > 1.0; `payout` > 1.0.
9. **Flag outliers in Y (do not exclude in the base case).** Compute Q1, Q3 and IQR = Q3 − Q1 of Y. Flag a company as an outlier if Y > Q3 + 3 × IQR or Y < Q1 − 3 × IQR.
10. **Run each model and check sample size.** Run steps 11–18 once for each model in section 1A step 5 (sheets `Regression_PE`, `Regression_PBV`, `Regression_PS`, `Regression_EVEBITDA`), on the whole cleaned European dataset. Each model needs at least 150 companies after cleaning. For the sector-dummy version, add one 0/1 column per GICS sector except a base sector, to the right of the X columns; each dummy's coefficient is that sector's premium or discount versus the base, as with the emerging-market dummy in Lecture 5, slide 17. The reversed LINEST column order still applies. Then choose each sample company's valuation multiple as in section 1A, step 6.
11. **Build `Correlations` and the VIF block** (section 3.3).
12. **Build `Regression`** (section 3.2).

13. **Recalculate and check for errors.** Formulas written by a library are not calculated until the file is opened. Recalculate the workbook (Excel, or LibreOffice in headless mode) and confirm that no cell outside the documented #N/A positions of LINEST shows `#N/A`, `#VALUE!`, `#DIV/0!`, `#REF!` or `#NAME?`.
14. **Cross-check independently.** Re-estimate the same regression outside Excel (for example Python with numpy least squares or statsmodels OLS, using the same rows). Every coefficient and R^2 must match the workbook to a relative difference below 1e-6. If they do not match, stop and find the cause (most often: reversed LINEST column order or misaligned rows).
15. **Run the diagnostics** in section 5 and write every result and flag to `Diagnostics`.
16. **Build `Valuation`** (section 3.4) for the sample companies only (`is_sample` = TRUE), and the scatter chart with all companies.
17. **Robustness runs.** Add a sheet `Robustness` after `Valuation`. Run it only when triggered:
    - If any outlier was flagged in step 9: re-estimate without the outliers and report how the coefficients, R^2 and each company's classification change.
    - If any VIF > `vif_severe`: re-estimate without the X variable with the highest VIF and report the same comparison.
    - The base case in `Valuation` always remains the full configured model on all valid companies.
18. **Value targets** (optional, section 3.5) if the user supplied companies outside the sample.
19. **Write `valuation_summary.md`** following section 6.
20. **Run the QA checklist** in section 8. Deliver only when every item passes.

5. Diagnostics and decision rules

Run every check below, write its value and result to `Diagnostics`, and apply the action exactly. A failed check never changes the base-case formulas; it changes what the report is allowed to conclude.

#	Check	Computed as	Threshold	Action
D1	Model significance	`Sig_F`	< 0.05 passes	Fail: state in the report headline that the model has no statistically significant explanatory power and that the classifications must not be used as valuation conclusions
D2	Explanatory power	`R2`, `Adj_R2`	R^2 < 0.20 = warning	Warning: report that error bands are wide and classifications are indicative only. For reference, the course's sector regressions have R^2 of 43.5–66% (Lecture 5, slides 17, 29, 38); the 0.20 threshold is our rule
D3	Coefficient significance	p-value per X	< 0.05 = significant	Not significant: keep the variable (it is theory-specified), list it as not significant in the report
D4	Coefficient sign	Sign OK column	Must equal the expected sign	Wrong sign: flag it, but do not treat it as an error by itself; the course's own regressions show signs that differ from theory (section 2, Lecture 5 slides 33 and 39). Check D5 first; if VIF is high, attribute to multicollinearity. If the wrong sign is also significant, put it in the report's "Key warnings"

#	Check	Computed as	Threshold	Action
D5	Multicollinearity	Pairwise `corr_X`, VIF	|r| > 0.80 or VIF > 5 = warning; VIF > 10 = severe	Warning: report that individual coefficients are unstable but predicted multiples remain usable. Severe: run the robustness model in step 17
D6	Negative predictions	Predicted multiple ⩽ 0	Any	Mark the company "Not valued (negative prediction)"; explain that the linear model is outside its valid range for that company
D7	Outliers in Y	Step 9 rule	Any flagged	Run the robustness model without outliers (step 17)
D8	Extreme residuals	|standardized residual|	> 3	Flag "check input data" for that company; do not exclude it
D9	Sample size	n vs k	n < 10 × k	Warning in the report
D10	Extrapolation (targets only)	X outside sample MIN–MAX	Any	Flag the target's valuation as extrapolated and less reliable

How to read a classification. The standardized residual measures how far the actual multiple sits from the fitted line, in units of SEy:

Standardized residual	Classification	Meaning
< −2.0	Significantly undervalued	Trades well below what its fundamentals imply
−2.0 to < −1.0	Moderately undervalued	Below, but within normal dispersion

Standardized residual	Classification	Meaning
−1.0 to +1.0	Fairly valued	In line with fundamentals
> +1.0 to +2.0	Moderately overvalued	Above, but within normal dispersion
> +2.0	Significantly overvalued	Trades well above what its fundamentals imply

About 95% of residuals are expected inside ±2 SEy if the model is well specified, so only the "Significantly" categories are strong signals. These are approximate bands based on SEy; they ignore the extra uncertainty from estimating the coefficients.

6. Output specification

`valuation_summary.md` must follow the template below: same headings, same order, every placeholder in `<>` filled. Numbers in the report must equal the workbook values after recalculation.

Formatting rules for the report:

- Multiples to 1 decimal (24.2x). Coefficients, standard errors and t-stats to 3 significant figures. p-values to 3 decimals, or "< 0.001".
- Rates as percentages with 1 decimal (15.0%), even though the workbook stores decimals.
- Translate each coefficient into plain words with the right unit. Because growth and payout are stored as decimals, the effect of **+1 percentage point** = coefficient × 0.01. For beta, report the effect of **+0.1** = coefficient × 0.1.
- Do not add conclusions that the diagnostics forbid (section 5).

Template:

```
# Relative valuation by regression — <sample name>, <date>

## Headline
<2–3 sentences: model quality in one line (R², Sig F), the most undervalued
and most overvalued companies with their standardized residuals, and any D1
failure.>

## Data
- Companies received: <N_raw>; used: <n>; excluded: <N_excl> (see table
```

```
below)
- Multiple: <Y>; earnings basis: <trailing/forward>; data date: <date>;
source: <source>
- Unit normalizations applied: <list or "none">

| Excluded company | Rule | Reason |
|---|---|---|

## Fitted model
<Y> = <b0> + <b_growth> × growth + <b_beta> × beta + <b_payout> × payout

| Variable | Coefficient | Std. error | t-stat | p-value | Expected sign |
Sign OK |
|---|---|---|---|---|---|---|

- R² = <R2>; adjusted R² = <Adj_R2>; SEy = <SEy>; F = <F> (Sig F = <Sig_F>);
n = <n>; df = <df>

**What the coefficients mean**
- +1 pp of expected growth → <Y> changes by <b_growth × 0.01>, holding beta
and payout fixed
- +0.1 of beta → <Y> changes by <b_beta × 0.1>, holding the others fixed
- +1 pp of payout → <Y> changes by <b_payout × 0.01>, holding the others
fixed

## Diagnostics
| Check | Value | Result |
|---|---|---|
<one row per D1–D10>

## Valuation results (sorted from most undervalued to most overvalued)
| Rank | Company | Actual <Y> | Predicted <Y> | Std. residual | % vs
predicted | Implied price | Current price | Upside | Classification | Flags |
|---|---|---|---|---|---|---|---|---|---|---|

## Robustness
<Only if step 17 ran: what changed in coefficients, R² and classifications.
List every company whose classification changed. Otherwise: "Not triggered.">

## Key warnings
<Bullet list of every failed or warning check, in plain words. "None" if
empty.>
```

```
## Limitations
- The model explains <R2 as %> of the cross-sectional variation in <Y>; the
rest
  comes from factors not in the model.
- Results reflect the market's pricing at <date>; coefficients change over
time.
- A low multiple versus prediction is an indication to investigate, not proof
of mispricing.

## Sources
<Every link used, grouped by company, as required in section 9.
Course material cited as (Lecture X, slide Y).>
```

7. Rules, edge cases and stop conditions

Always:

- Use live formulas for every statistic and valuation in the workbook. Pasted values are allowed only in `Raw_Data`.
- Log every exclusion, normalization, flag and assumption. Nothing is removed or changed silently.
- Use the same rows, in the same order, in `Clean_Data`, `Regression`, `Valuation` and the independent cross-check.
- Keep the base-case model exactly as configured, even when a variable is not significant or has the wrong sign. Alternative models go only in `Robustness`.

Never:

- Invent, estimate or fill in a missing value (no averages, no interpolation, no values from memory).
- Drop a company because its result looks unusual. Only the rules in step 6 can exclude a company.
- Mix trailing and forward EPS, or betas from different sources or windows.
- Read LINEST coefficients assuming they are in the same order as the X columns.
- Report a valuation conclusion that a failed diagnostic forbids.

Stop conditions. Stop, deliver nothing except a short message to the user explaining the problem and what you need:

Condition	What to ask the user for
A required column is missing or cannot be mapped with certainty	The column, or confirmation of the mapping
n < `min_observations` after cleaning	More companies, or explicit permission to run on a small sample
EPS basis is mixed or unclear	Which basis to use, and data on that basis
A configured X variable has no expected sign in section 2	Its expected sign
The workbook and the independent cross-check disagree and you cannot find the cause	Nothing: report the mismatch and both sets of numbers
The workbook cannot be recalculated, so values cannot be verified	Nothing: report that the file is unverified

Edge cases:

- **All companies flagged as outliers in one direction** (for example an industry with structurally high multiples): do not exclude; mention it in the report.
- **A company with payout = 0** is valid; keep it.
- **Constant X column** (every company has the same value): LINEST cannot estimate it. Remove that X from the model, log it, and say so in the report.
- **Targets with missing X values:** do not value them; list them as "not valued — missing data".
- **A model with fewer than 150 companies after cleaning:** do not run it yet. Extend the universe to MSCI Europe IMI (section 1A, step 1); if it is still short, tell the user and run only the models that pass.
- **A sample company not eligible for any model** (for example negative earnings, book value and EBITDA): report it as "Not valued by regression" and tell the user to value it with comparables only.
- **Fiscal years not ending in December** (SSE: 31 March; Siemens Energy: 30 September): use the latest full fiscal year and flag it; never mix it silently with calendar-year data.
- **Parent and subsidiary both in the dataset** (for example EDP and EDP Renováveis): keep both, but flag it in the report, because their residuals are not independent.

8. Final QA checklist

Deliver only when every item passes. Copy this list into the `Log` sheet with PASS or FAIL next to each item.

- [ ] `Raw_Data` row count equals rows received; `Clean_Data` + `Excluded` = `Raw_Data`
- [ ] Every excluded company has a rule number and reason, and every figure in Clean_Data has a linked source (section 9)
- [ ] `growth` and `payout` are in decimals; any normalization is logged; EPS converted into the price currency wherever the two differ, with the FX rate logged; every replaced beta logged with its original value
- [ ] Y in `Clean_Data` is the formula price ÷ eps for every row
- [ ] Named ranges `Y_range` and `X_range` cover exactly rows 2 to n + 1
- [ ] Coefficients are read with the reversed LINEST order (X variable j in column k − j + 1)
- [ ] Workbook recalculated; no error values outside LINEST's documented #N/A cells
- [ ] Independent cross-check matches coefficients and R^2 to < 1e-6 relative difference
- [ ] Diagnostics D1–D10 all computed and written to `Diagnostics`
- [ ] Robustness runs done where triggered, or marked "Not triggered"
- [ ] Spot check: for 3 companies, recompute the predicted multiple by hand from the coefficients and confirm it equals column F
- [ ] `Valuation` ranks run from 1 (most undervalued) to n with no gaps except ties
- [ ] Every number in `valuation_summary.md` equals the recalculated workbook value
- [ ] The report draws no conclusion that a failed diagnostic forbids
- [ ] Both files delivered: `regression_valuation.xlsx` and `valuation_summary.md`

9. Source linking (mandatory)

Every figure and every judgment you use in this regression must be traceable through a link. A figure without a linked source counts as missing: exclude it (step 6, rule 2).

- **Link each figure.** For every value you collect (price, shares, EPS, DPS, beta, expected growth, payout, ROE, net margin, ISIN, FX rate, industry beta), record the full URL of the exact page or document it came from and the access date, in the `source_urls` column. Format: `field: URL (accessed YYYY-MM-DD)`, entries separated by `|`.
- **Link the exact page, not a home page.** For example the annual report PDF plus page number, or the exchange's quote page for that share.

- **Terminals without a public URL** (Bloomberg, Eikon): write the terminal, the ticker, the field or screen used, and the date.
- **Conflicting sources:** if two sources differ by more than 2%, link both and state which one you used and why.
- **Report:** end `valuation_summary.md` with a "Sources" section listing every link, grouped by company. Cite course material as (Lecture X, slide Y).
- **Never** cite a source you did not actually open, and never write a URL from memory.